

HEALTH DISEASE PREDICTION

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Abstract:-

Heart disease remains one of the leading causes of mortality worldwide. Early and accurate prediction can help clinicians provide timely diagnosis and reduce patient risk. This research presents a machine learning–based system for predicting the likelihood of heart disease using clinical parameters such as age, cholesterol level, chest pain type, resting blood pressure, fasting blood sugar, thalach, oldpeak, and more. Multiple machine learning models including Logistic Regression, Random Forest, Gradient Boosting, and a Stacking Ensemble were developed, trained, and evaluated. The proposed stacking classifier improves predictive performance compared to individual models. Results demonstrate the potential for ML-driven decision support in healthcare risk assessment.

1. Introduction

Cardiovascular diseases continue to cause high mortality and disability rates globally. Early-stage diagnosis is crucial as it allows better treatment and prevention. Machine learning offers powerful tools to analyze patient clinical datasets and identify patterns that can predict heart disease efficiently and accurately.

With digital healthcare systems generating large volumes of medical data, automated prediction tools can support clinical decision-making. In this study, we propose a heart disease prediction model using machine learning and ensemble learning to improve reliability and performance.

2. Literature Review

Several studies have explored machine learning algorithms for heart-disease prediction. Logistic Regression and Decision Trees have been used for interpretable models, while Random Forest, Support Vector Machines, and Gradient Boosting demonstrate improved performance with non-linear patterns. Recent research explores hybrid and ensemble approaches. However, existing models often lack reproducibility or strong evaluation, and comparatively few analyze results between baseline and ensemble models.

Research Gap Identified

- Existing systems often focus on single-model performance without ensemble comparison.
- Lack of standardized pipeline for preprocessing and reproducibility.
- Limited analysis of model generalization on smaller datasets.

3. Problem Statement

To develop an accurate and reliable machine learning model that predicts whether a patient is likely to have heart disease based on clinical features.

4. Objectives

- To preprocess and analyze heart disease datasets.
- To develop prediction models using different ML algorithms.
- To design a stacking ensemble model to improve classification performance.
- To compare accuracy, confusion matrix, and ROC metrics between models.
- To evaluate how preprocessing and feature scaling improve performance.

5. Proposed System

System Architecture

Dataset → Preprocessing → Feature Scaling → Train/Test Split →
Model Training → Evaluation → Ensemble Model → Results & Interpretation

Algorithms Used

Model	Description
Logistic Regression	Baseline linear classifier
Random Forest	Tree-based ensemble model
Gradient Boosting	Boosted ensemble for improved accuracy
Stacking Ensemble	Meta-model combining RF & GB with Logistic Regression

6. Dataset Description

The dataset consists of medical attributes including:

Feature	Description
Age	Patient age in years
Sex	1 = male, 0 = female
cp	Chest pain type
trestbps	Resting blood pressure
chol	Cholesterol in mg/dl
fbs	Fasting blood sugar > 120 mg/dl
thalach	Maximum heart rate
oldpeak	ST depression induced by exercise
target	Disease present (1) / absent (0)

Dataset size used: **303 records, 14 features** (synthetic or real dataset optional).

7. Preprocessing

- Missing value handling using median imputation.
- Standard scaling of numeric features.
- Stratified train-test split (80/20).
- Removal of noise and normalization.

8. Model Development & Training

Steps:

1. Load dataset
2. Preprocess and scale features
3. Train baseline models
4. Train stacking ensemble

5. Save best model
6. Generate evaluation graphs

9. Evaluation Metrics

Metrics used to compare performance:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion matrix

10. Results & Discussion

The Stacking Ensemble achieved better performance than individual classifiers.

Example result format (replace with actual output after run):

Model	Accuracy
Logistic Regression	87%
Random Forest	91%
Gradient Boosting	92%

Stacking Ensemble (Proposed) 94%

11. Conclusion

The proposed stacking ensemble model improves prediction accuracy for heart disease compared to baseline models. The system demonstrates that machine learning can support clinical judgment and guide early diagnosis. Future work may include deep learning approaches and real-time deployment in hospitals.

12. Future Scope

- Real medical dataset evaluation
- Integration into web or mobile healthcare applications
- Explainable AI methods (SHAP/LIME)
- Cloud deployment & IoT health monitoring